Approve the clear refunds automatically.
Send the rest to the right person, with reasons.
Let Refund Ops preview rule changes before they go live.

Smart Refunds

PYTHON · STREAMLIT · CLAUDE HAIKU 5.5 & OPUS 5.5

01/	Where the work breaks
02/	The automatic line, and the evidence
03/	Reviewers and the manager
04/	Measuring, and where AI is used
05/	Deferred, and what comes next

Where the time and money go

DECISIONS	REVIEW TIME	WAIT	CHARGEBACKS
137/248	**1,120 min**	**1.6 days**	**$2,519**
decisions made by people	of review; 40% on never-arrived claims	median wait, even for a $20 refund	in 8 chargebacks; 7 had the red flag in the notes

BEFORE	Request → old rule (small amount, older account, no prior refunds; ignores notes, lets missing data pass) → one shared queue → a person reads every case
AFTER	Request → clean the data → read notes into facts → ordered rules → **auto-approve, or the right lane with reasons**

Source: history.csv, 248 decisions, June–August 2026

What the system decides alone

AUTO-APPROVE ONLY IF EVERY CHECK PASSES

- Up to $100, account 9+ months old
- No refund in 12 months; fraud flag an explicit “no”
- Evidence quoted from the agent’s note: photo for damage over $50, picking error for wrong items from $55
- Change of mind within 14 days; change of mind and late delivery under $55 only
- Nothing in the notes contradicts the claim

	Old rule	New rules
Auto-approved	111	103
Money approved	$3,377	$3,303
Bad outcomes approved	1	1
Bad data approved	4	0
Damage >$50, no photo	3	0

REPLAYED ON 248 PAST DECISIONS

ALWAYS A PERSON

Never-arrived claims, any fraud signal, bad data, duplicates, notes that try to instruct the system. Rejections are only drafted.

THE ONE MISS, IN BOTH

A $49 claim with empty notes.

CAP AT $200

+6 approvals, no new bad outcomes. At $300 a $210 liquid-damage claim slips through.

The right queue, with the reasons

1/ WHERE IT LANDS

- Five lanes: Fraud team, Delivery disputes, Damage & evidence, Policy rejections, Other
- Related cases grouped: Northfield depot, 5 cases, $726
- Waiting tab for a photo or carrier answer

2/ WHAT THEY SEE

- Why the rules sent it here
- Flags from the notes, each with its quote
- The original notes
- Similar past cases and outcomes
- A warning if a note tries to instruct the system

3/ WHAT THEY DO

- Approve in full or part, reject with a drafted message, ask for evidence, or move lane
- Overriding needs a reason code
- Logged with the rule version; refunded at most once

EXAMPLE

RF-2181, $1,150, carrier signature on file. Lands in Delivery disputes with a drafted rejection, next to two similar past approvals that became chargebacks.

Change a rule, see the effect first

1/	2/	3/	4/
Describe it in a sentence	Confirm what it means	Preview on history and today's queue	Publish with a reason; roll back any time

"Stolen-card cases go straight through."

Asks: confirmed by the fraud team, or only reported?

Preview ("reported"): +11 auto-approvals in history, 151 reviewer minutes saved, no new bad outcomes; RF-2182 and RF-1070 approved today

Warns: skips 4 checks; largest new approval $610; only 11 past cases

"Anything over $300 comes to us."

Preview: no effect today. Names the rules that already catch these and points to the $100 cap.

Still a useful guard if the cap is raised.

Rules run in order, first match decides; four locked safety rules come first.

How we'd know it works, or doesn't

BEFORE SWITCH-ON

- Replay on past decisions (done)
- Labelled note-reader eval (done)
- Two weeks of shadow mode, comparing every difference (next)

WATCH WEEKLY

Number	Good looks like
Share auto-approved	**about 40%**
Chargebacks on auto-approvals	**0**
Overrides per lane	**<10%**
Audit of 20 auto-approvals	**0–1 disagree**
Wait for a person	**<1 day**

ROLL BACK WHEN

- Any chargeback on an auto-approval
- Audit disagreement above 5%
- A flag spike or an unreviewed incident

Targets to agree with Refund Ops. If the note reader is down, those requests go to people: the risk is queue size, not money.

Where AI is used, and where it isn’t

USED FOR

- Reading notes in any language into fixed facts, each with its quote (Haiku 5.5)
- Drafting a rule from the manager’s sentence, which she confirms (Opus 5.5)

NOT USED FOR

The approve decision, thresholds, data cleaning, duplicates, paying once, place-name spellings, preview numbers. These stay testable and safe when the model is down.

THREE MODELS, 43 LABELLED CASES

Model	Red flags	Unsafe	Cost
Haiku 5.5	100%	0	0.05¢
Sonnet 5.5	94%	0	0.95¢
Opus 5.5	94%	0	1.9¢

Equal on safety, so the cheapest won: about $0.15 a week at 300 requests. Same outcome as Opus on all 248 past decisions at the $100 cap.

Above $200: Opus flags a liquid-damage claim Haiku misses. Raising the cap needs a new eval first.

Deferred, and what I'd do next

DEFERRED ON PURPOSE

- Real payments, messages and sign-in (simulated)
- Bulk decisions on an incident
- Auto-expiry of temporary rules
- Shadow mode and monitoring: designed, not built
- Reviewer corrections feeding the eval set
- Rule drafting failed 1 of 5 live tries; falls back safely, now logged

NEXT

1/ Two weeks of shadow mode on live requests

2/ Carrier tracking, so "never arrived" is decided on facts

3/ Weekly: reviewer corrections become new labelled cases

4/ Rollback triggers wired to alerts

5/ Revisit the $100 cap with shadow data, after adding RF-2082-type cases to the eval

6/ Other model providers, judged by the same eval and backtest